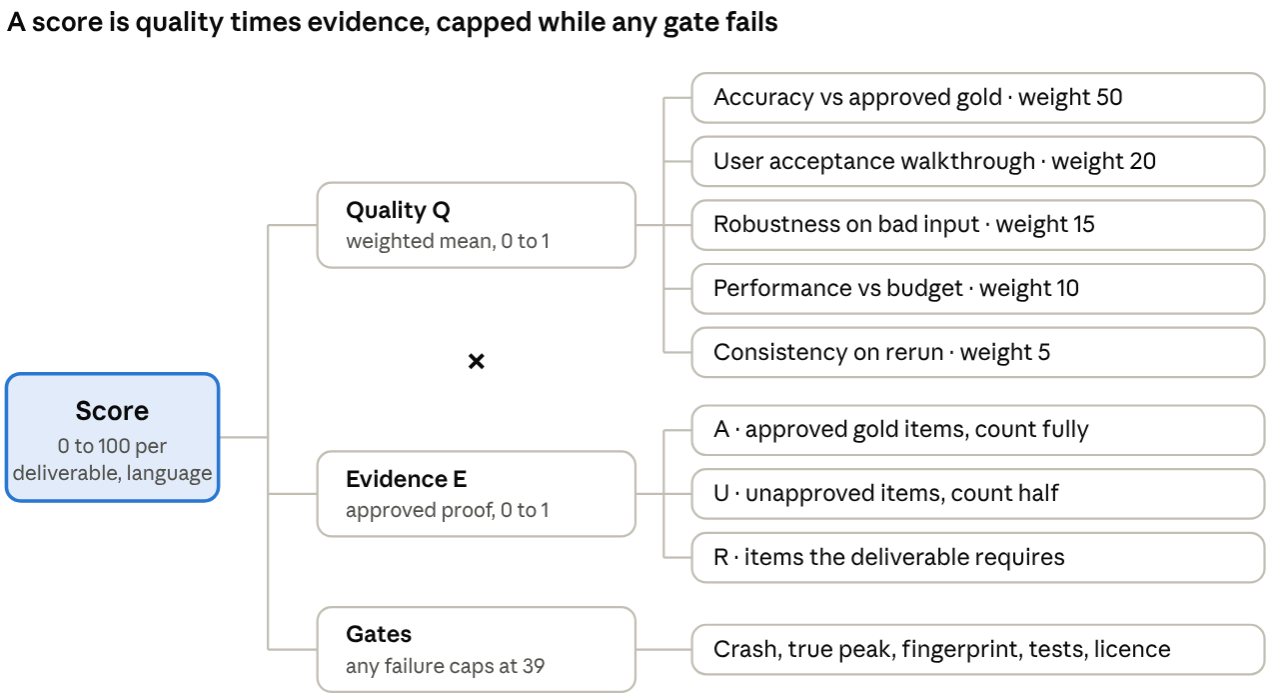
Audio Reverse Engineering — Deliverable Scorecard

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Every deliverable gets one score from 0 to 100 per language, recomputed on every scoring run. A score of 100 means *done* in the Definition of Done. The same formula applies to every feature, every instrument family and every language, so progress can be compared and tracked over time.

How a score works

A score multiplies how good the deliverable is (**quality**) by how much approved proof stands behind that measurement (**evidence**). It's capped at 39 while any hard check fails.



score composition · quality × evidence, gate cap

$$\text{Score}(d, \ell) = \left\lfloor 100 \cdot Q(d, \ell) \cdot E(d, \ell) \right\rfloor, \quad \text{capped at } 39 \text{ if any gate fails}$$

$$Q = \frac{\sum_c w_c \, n_c}{\sum_c w_c}, \qquad n_c = \operatorname{clamp}\!\left(\frac{x_c - \text{floor}_c}{\text{target}_c - \text{floor}_c},\ 0,\ 1\right)$$

$$E = \min\!\left(1,\ \frac{A + 0.5\,\min(U, R)}{R}\right)$$

Here d is the deliverable and ℓ the language (“any” for deliverables that don’t depend on language). Each component c has a raw measurement x, a floor that scores 0, a target that scores 1, and a weight w. For lower-is-better metrics, such as word error rate, the floor is set above the target and the same formula works. A is the number of approved gold items measured, U the number of unapproved items, and R the number the deliverable needs.

A deliverable reaches 100 only when every component is at or past its target, every required gold item is approved, and no gate fails. Better-than-target results don’t raise the score above 100 or make up for a weak component elsewhere.

Components

Each deliverable uses the components that apply to it. A missing component isn’t counted as zero; it’s left out of the weighted mean. Default weights are shown below; a deliverable can override them in the registry, as long as the weights still add up to 100.

Component	Measures	Raw value x	Floor (scores 0)	Target (scores 1)	Default weight
Accuracy	How close output is to approved gold	The deliverable’s metric: F-measure, word error rate, boundary error, SDR, onset F, and so on. Several metrics are averaged after normalising	Gate 0 baseline, or the “unusable” value when there is no baseline	The DoD target	50
User acceptance	Whether the user can do the action	Share of walkthrough steps passed, on 2 songs per language	0%	100%	20
Robustness	Behaviour on bad and odd input	Share of edge-case tests passed: corrupt, silent, clipped or very long files, extreme tempo, noise	0%	100%	15
Performance	Speed and memory	Wall clock and peak VRAM against budget	3× budget	At or under budget	10
Consistency	Same input, same output	Share of reruns identical (cloud results replayed from cache)	0%	100%	5

Gates (no weight, but any failure caps the score at 39):

- A crash or stack trace on valid input
- True peak above −0.7 dBFS on any render
- An unintended render-fingerprint change
- A failing hermetic test in the deliverable's area
- A dependency or checkpoint outside the licence allowlist

Evidence and trust

A score only counts as much as the approved data behind it. Evidence E is what turns a good-looking number into a trusted one.

- **Gold item.** One song or clip with an approved reference for what the deliverable measures: notes, word-timed lyrics, drum hits, sections or stems.
- **Approval record.** Each gold item stores its author, the approving professional musician (from an approved list), the date, the scope (notes, lyrics, timing) and a checksum of the files. If a file changes after approval, its approval no longer applies.
- **Unapproved items** count half, but only unreviewed candidates with validity $V \geqslant 85$ qualify. They let us track progress before musicians deliver, but they can't take a score past 50.
- **Required amounts** R, per language and per deliverable (proposed):

Candidates are found, validity-scored and queued for review by the background loop in Gold Data Sourcing.

Evidence set	Required per language	Applies to
Songs with word-timed lyrics and melody	8, across ⩾ 2 genres	Voice deliverables (FR-12 to FR-14, V1–V11)
Clips per instrument family	3	FR-4, FR-11 and each family; not language-specific
Songs with sections and chords	5	FR-5, U9
Walkthrough songs	2 (1 gold, 1 not)	Every user action (U, V)

Language trust score. For a language ℓ, it's the mean score of all language-dependent deliverables, with Must items weighted twice. It answers one question: how far can we trust what the app says about audio in this language? Next to it the report shows the language's evidence coverage E, so a low score is easy to tell apart from missing data.

Uncertainty. Each accuracy value is reported with a 90% bootstrap interval over gold items. The score uses the point value; a wide interval is a signal to add evidence, not a different score.

Score and performance

A sheet sets out how the music should go; a recording is one performance of it. Singers push and pull the beat, swing straight eighths, add slides and ornaments, and hold notes longer or shorter than written. Neither is wrong, so every song carries two layers, and each check is scored against the layer it is about.

Layer	Holds	Time unit	Truth comes from	Checks it feeds
Score	Notes, written durations, meter, chords, voices, lyric syllables per note	Bars and beats	Approved sheet music	Score-level note F, chords, syllables on notes, notation
Performance	When each note and word actually sounds; pitch curve; loudness	Seconds	The recording	Word highlighting, word-boundary error, onset error, playback sync
Tempo map (the link)	The second each beat falls on, plus each note's offset from its beat	Beat ↔ second	Score aligned to the recording	Places every score item in the recording, and back

Timing truth for songs that only have an approved score (the three Lithuanian songs today):

1. Align the approved score to the recording in batches, unit by unit (see Units and Batch Alignment), voice by voice on separated stems, with a score-follower kept separate from the app's own aligner.
2. A musician spot-checks anchor points (every phrase start, at least one per 4 bars, plus 10 unannounced random points).
3. After the spot-check passes (≥ 95% of points within 50 ms, after corrections), derived timings count **fully for bars and above**, and stay at **half for beats, notes and words**. Full timing evidence below bar level comes only from songs a musician has timed completely by ear. Each language needs at least one such benchmark song.

So the three Lithuanian songs count fully for score-level checks, half for bar-level timing until spot-checked, and half for note and word timing until a musician times them by ear. Because they are jazz swing with improvised solos, note-level checks run only on their written and embellished units.

Expression or error? Comparing a transcription with the score layer:

Difference from the written score	Counts as
Onset off the written grid but consistent with the tempo map	Expression, not an error
Swing or dotted feel on written straight notes	Expression; reported as a swing ratio
Grace notes, slides, scoops, vibrato not written	Expression, unless held as a new pitch for ⩾ 1 beat
A different held pitch, a missing note, an extra note	Error
A word or note sung differently from the sheet, consistently	Performed variant: flagged for a musician, who can mark it “as performed” so the gold keeps both versions

Languages

English and Lithuanian run as parallel tracks from now on, and both are scored from the first run. Every language is a *language pack*: the scoring formula stays the same, and the pack supplies what is specific to the language.

Pack part	English	Lithuanian
Text normalisation for word error rate	Lower-case, strip punctuation, spell out numbers	Same, but keep diacritics (ą č ę ė į š ų ū ž), since they change meaning
Syllable rules	Dictionary-based hyphenation	Existing `split_into_syllables_lt` in `vocal_transcriber.py`, checked against gold
Aligner model	English acoustic model	Multilingual (MMS) or a Lithuanian-trained model

Pack part	English	Lithuanian
STT / TTS candidates	Cloud services, Whisper large-v3 / turbo	Cloud services with Lithuanian support, Whisper, a Lithuanian-tuned model; Edge-TTS `lt-LT` voices
Gold set today	0 items	3 songs (*Grikiai*, *Virusas*, *Prie Topolio*), provided and approved by musicians
Evidence coverage *E* today	0	Score-level checks 0.38 (3 approved of 8); bar-level timing 0.19 until spot-checked; note and word timing 0.19 until songs are timed by ear

Adding a language means adding a pack and a gold set; the code and formula stay as they are. Its scores start at 0 and rise as evidence and quality arrive.

Fastest trust gain now: English. Lithuanian stands at 0.38 for score-level checks with its three approved songs (word and note timing need songs timed by ear by a musician) and needs 5 more; English has none, so it gets sourcing and commissioning priority.

Bands and rules

Score	Band	Meaning
0–39	Not trusted	A gate is failing, or there's too little quality or evidence to rely on
40–69	Developing	Works on some material; not to be relied on for real songs
70–89	Usable	Reliable on most material; known gaps remain
90–99	Release candidate	Meets nearly every target with approved evidence
100	Done	Every component at target, full approved evidence, no gate failing

Rules that keep the score honest

1. **Machines measure, people approve.** Every component except user acceptance is computed by the scoring run. Walkthroughs are recorded with screenshots or video, and the record is linked from the run.

2. **Targets and floors are frozen per release.** Changing one needs a recorded decision with the reason, and old scores are recomputed against the new values so trends stay comparable.
3. **Gold data is never trained on.** Gold items are held out from any model training or tuning, and the scoring run checks their checksums against a training manifest.
4. **Regression blocks merges.** CI fails when any deliverable drops more than 2 points, or when a language trust score drops at all, against `main`.
5. **No rounding up.** Scores are floored to the integer below, so 99.6 shows as 99.
6. **Release rule.** A release ships when every Must deliverable it includes scores 100 in each of its languages. Should and Could items ship at ⩾ 70 or are cut.

Worked example

V1, word highlighting in Lithuanian, scores **12** on today's evidence. Quality is moderate, but V1 is a timing check: the 3 approved songs only have derived timings, so they count half (1.5 of the 8 needed). *All quality measurements below are illustrative; real ones come from the first scoring run.*

Component	Raw value	Floor → target	Normalised	Weight
Accuracy: word-boundary error	170 ms	300 → 80 ms	0.59	50 (averaged with the row below)
Accuracy: word error rate	34%	40% → 28%	0.50	
User acceptance	7 of 10 steps	0 → 100%	0.70	20
Robustness	18 of 20 cases	0 → 100%	0.90	15
Performance	12 min	30 → 10 min	0.90	10
Consistency	10 of 10 reruns identical	0 → 100%	1.00	5

Quality Q = (50 × 0.545 + 20 × 0.70 + 15 × 0.90 + 10 × 0.90 + 5 × 1.00) / 100 = 0.6875. With evidence E = 1.5/8 = 0.1875, Score = ⌊100 × 0.6875 × 0.1875⌋ = ⌊12.9⌋ = 12.

- A musician times one of the three songs completely by ear: E = (1 + 0.5 × 2) / 8 = 0.25, so the score becomes **17**. A spot-check alone doesn't raise it, because V1 is a word-level check.

- All three timed by ear: E = 0.375, score **25**. Eight timed songs: E = 1.00, score **68**, with no code change.
- After that, only better quality moves it: the path to 100 is the per-deliverable plan we write next.
- The same deliverable in English scores 0 until its first approved songs arrive, while its Q is already shown.

Deliverable registry

Every item below is scored. User actions (U, V) are what the user experiences; requirements (FR) and instrument families (F) are the engineering underneath. Language-dependent items get a separate score for EN and for LT. The next step is to write, for each row, what it takes to reach 100.

ID	Deliverable	Scored for	Main accuracy metric	Evidence set
FR-1	Ingest	any	Manifest completeness	Walkthrough songs
FR-2	Separation	any	Vocal SDR; lead/backing leakage	Gold reference stems
FR-3	Pitch	any	Raw pitch accuracy (±50 cents)	Family clips
FR-4	Transcription	any	Note F-measure	Family clips
FR-5	Structure	any	Section boundary F; chord accuracy	Sections and chords songs
FR-6	Synth matching	any	MR-STFT win rate	Held-out synth clips
FR-7	Session assembly	any	Batch success rate	Walkthrough songs
FR-8	Score view	any	Cursor sync error	Walkthrough songs
FR-9	Evaluation	any	Metric coverage of gold items	All sets
FR-10	Interfaces	any	Contract tests passed	n/a (tests)
FR-11	Instrument families	any	Family label accuracy	Family clips
FR-12	Voice understanding	EN, LT	WER; boundary error; syllables on note	Lyric songs

ID	Deliverable	Scored for	Main accuracy metric	Evidence set
FR-13	Lyric editing	EN, LT	Round-trip fidelity	Lyric songs
FR-14	Re-voicing	EN, LT	F0 spread vs original	Lyric songs
FR-15	Gold provenance	any	Share of gold with valid approval	All sets
F-piano	Piano / keys	any	Note F; hand-split accuracy	3 piano clips
F-guitar	Guitar	any	Note F; playable TAB share; chord names	3 guitar clips
F-bass	Bass	any	Octave errors; onset error	3 bass clips
F-drums	Drums and kick	any	Kick, snare, hi-hat onset F	3 drum clips
F-synth	Synth lead / pad	any	Note F	3 synth clips
F-sfx	SFX / other	any	Event onset F	3 SFX clips
U1–U18	User actions (see DoD)	any; U5, U6, U8 per family	Walkthrough + the linked FR metric	Walkthrough songs
V1–V11	Voice actions (see DoD)	EN, LT	Walkthrough + FR-12/13/14 metrics	Lyric songs + walkthrough songs
FR-16	Gold data sourcing	EN, LT	V calibration (share of V ⩾ 85 approved); gap fill rate	Musician review outcomes
R1–R10	Review actions (see Gold Data Sourcing)	any	Walkthrough	Walkthrough candidates
FR-17	Score and performance layers	any	Beat error vs spot-checked anchors; written durations recovered by quantisation	Spot-checked anchors on gold songs

ID	Deliverable	Scored for	Main accuracy metric	Evidence set
FR-18	Batch alignment (unit ladder L0–L10)	any	Route accuracy; boundary error per level; chord-span accuracy; note F on written units only	Musician-marked routes and modes; benchmark songs
FR-19, M1–M8	Loops and mixes from units	any	Click-free loop rate; compatibility ranking judged by listeners; walkthroughs	Walkthrough songs; listening panel

Tracking mechanism

The scorecard is code in `apps/remix_studio/scorecard/`, built in Phase 0 before any feature work, so every later change is measured from day one.

Part	Where	What it does
Registry	`scorecard/registry.yaml`	Every deliverable: components, metrics, floors, targets, weights, evidence set, languages. Frozen per release tag
Language packs	`scorecard/languages/en.yaml`, `lt.yaml`	Normalisation, syllable rules, aligner and engine choices, required evidence
Gold approvals	`approval.json` next to each gold item	Author, approving musician, date, scope, SHA-256 of the files
Metric providers	One `measure()` per FR and family	Returns raw values per gold item; no scoring logic inside
Scoring run	`uv run python -m scorecard run [--only FR-12] [--lang lt]`	Runs providers, applies gates, computes Q, E and score

Part	Where	What it does
History	SQLite `data/remix/db/remix_studio.db`, tables `score_runs` and `scores`	One row per run, deliverable, language and component, with commit hash; never overwritten
Walkthroughs	`scorecard/walkthroughs/<release>/<id>-<lang>.json`	Step results plus a link to the recording
Report	`uv run python -m scorecard report`	Quality Q, evidence E and score, each shown separately, with change since last run and since `main`, language trust scores, and the weakest component of each deliverable
CI	Hermetic tier + a nightly run on the workstation GPU	Hermetic: checks the scoring maths on stub data. Nightly: real models and services, regression rule applied

Done when (the scorecard is itself a deliverable, and must score 100 first):

- [ ] The scoring maths has hermetic tests, including this doc's worked example (12 now, 17 with one song timed by ear, 68 with 8)
- [ ] Every registry row has a metric provider, even if the provider only reports "no evidence"
- [ ] The first nightly run stores a baseline for every deliverable in EN and LT
- [ ] The report shows each deliverable's score, its trend and its weakest component
- [ ] A deliberate regression in a test branch makes CI fail

Once the baseline exists, we go through each registry row and write its path to 100.